Trade, Natural Resources, and the Environment

A fully derived study companion: Comparative Advantage under Climate Change, Technology Adoption, Structural Transformation, Selection and Agricultural Productivity Gaps, Endogenous Food Policy, and the Food Problem
with lesson quizzes and a final exam

Study notes — Part V of the Trade Theory Companion Series

Contents

Part I

Climate Change and Comparative Advantage in Agriculture

This part is based on Costinot, Donaldson, and Smith (2016, *JPE*), "Evolving Comparative Advantage and the Impact of Climate Change in Agricultural Markets: Evidence from 1.7 Million Fields around the World." The paper asks how much of the damage from climate change to global agriculture can be undone by two margins of adjustment: farmers switching *crops* within a country, and consumers switching *trading partners* across countries. Both margins operate through the same underlying force — comparative advantage — so the paper builds a Ricardian trade model with a continuum of heterogeneous fields and estimates it on a novel, extremely granular productivity data set.

1 Lesson 1: Motivation, Data, and the Fréchet Model of Field-Level Yields

1.1 Two margins of adjustment

Intuition: The Philippines and rice

Rice accounts for roughly a third of Filipino agricultural land, output, and trade value. Suppose climate change lowers the productivity of rice specifically in the Philippines. Two margins can cushion the blow: (1) **production** — farmers reallocate land to other crops whose local productivity is less affected; and (2) **trade** — Filipino consumers import more rice from countries where rice productivity fell by less. Both margins are powered by the same object: how a country's productivity in one crop compares, in *relative* terms, to its productivity in other crops and to other countries' productivity in the same crop. That comparison is exactly comparative advantage.

1.2 FAO–GAEZ: measuring agro-climatic potential at the field level

Definition: FAO-GAEZ potential yields

The Food and Agriculture Organization's Global Agro-Ecological Zones (FAO-GAEZ) project reports **potential yields** (tonnes/hectare) for many crops on a fine grid of cells ("fields") covering the world's land surface. Potential yield is an agro-climatic construct — combining soil type, elevation, land gradient, rainfall, temperature, humidity, wind, and sun exposure — for the period 1961–1990 and under climate-change projections for 2071–2100, under scenarios such as rain-fed/irrigated and low-input/high-input farming. Crucially, potential yield is defined as the yield *if the entire field were devoted to that one crop*: it is not what is actually grown, since farmers optimally specialize plots according to comparative advantage.

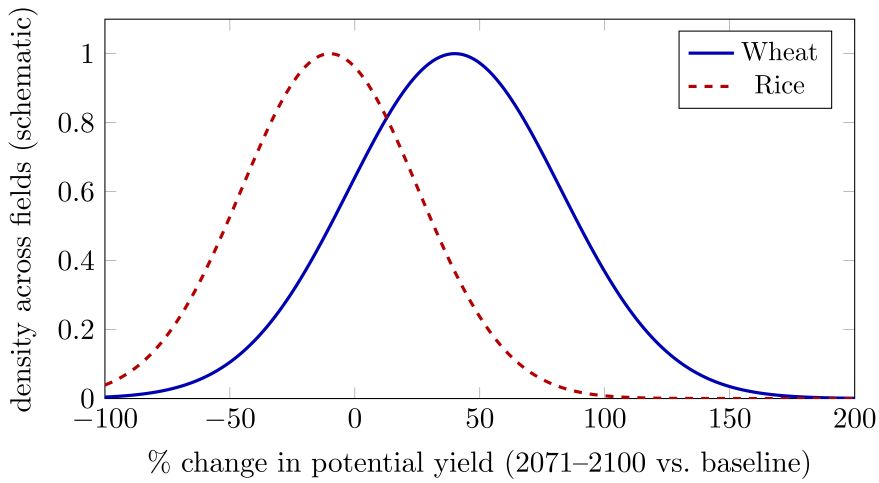

Figure 1: Schematic version of CDS's Figure 1: predicted yield changes are extremely heterogeneous across fields *and* across crops within the same field — some fields gain, others lose, even for the same crop. This heterogeneity is what production-side adjustment exploits.

1.3 Environment

Definition: Setup

Countries $i, j = 1, \ldots, N$; crops $k = 1, \ldots, K$ plus an outside good (manufacturing and services, produced with labor only). Country i has L_i units of land and N_i units of labor. Land is organized into fields $f \in F_i$, and each field is itself a continuum of heterogeneous **plots** $\omega \in [0, 1]$. Technology is constant returns to scale; markets are perfectly competitive; trade in each crop is subject to iceberg costs $d_{ij,k}$ (with $d_{ii,k} = 1$).

Definition: Preferences

Quasi-linear utility over the outside good C_n^0 and a CES/CES-nested agricultural aggregate:

$$U_n = C_n^0 + b_n \ln C_n, \qquad C_n = \left[\sum_k (b_{n,k})^{1/\kappa} C_{n,k}^{(\kappa-1)/\kappa}\right]^{\kappa/(\kappa-1)}, \qquad C_{n,k} = \left[\sum_i (b_{ni,k})^{1/\sigma} C_{ni,k}^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)}.$$

Here κ is the elasticity of substitution *across crops* (e.g. wheat vs. corn) and $\sigma > \kappa$ is the (typically larger) elasticity of substitution across *origin countries* for the same crop — an Armington structure layered on top of cross-crop CES.

Derivation: Field-level technology and the Roy–Fréchet plot problem

Labor produces the outside good linearly, $Q_i^0 = A_i^0 N_i^0$, which pins down the wage $w_i = A_i^0$ under free entry and a numeraire normalization. On the land side, plot ω in field f of country i, if planted with crop k, yields

$$Q_{i,k}^f(\omega) = A_{i,k}^f(\omega)\, L_{i,k}^f(\omega),$$

where $A^f_{i,k}(\omega)$ (land productivity) and $N^f_i(\omega)$ (labor intensity) are independent Fréchet draws with shape θ and means $\mathbb{E}[A^f_{i,k}(\omega)] = A^f_{i,k}$, $\mathbb{E}[N^f_i(\omega)] = v_i$. Each plot chooses the best of $K+1$ options (grow one of the K crops, or leave the plot idle and supply the implicit labor value $w_i N^f_i(\omega)$):

$$\max\left\{p_{i,1}A^f_{i,1}(\omega),\ \ldots,\ p_{i,K}A^f_{i,K}(\omega),\ w_i N^f_i(\omega)\right\}.$$

This is exactly the Roy–Fréchet discrete-choice structure from Lesson 8 of the Ricardo-to-EK companion, now applied to *land* rather than labor.

Derivation: Closed-form land shares and output

Fréchet aggregation delivers the field-level share of land allocated to crop k,

$$\boxed{\pi^f_{i,k} = \frac{(p_{i,k}A^f_{i,k})^\theta}{(\alpha_i)^\theta + \sum_{k=1}^K (p_{i,k}A^f_{i,k})^\theta}}, \qquad \alpha_i \equiv \frac{A^0_i}{v_i},$$

and, using the truncated-mean property of Fréchet (conditional productivity exceeds unconditional productivity because only favorable draws are planted),

$$Q^f_{i,k} = L^f_i\ A^f_{i,k}\left[\frac{(p_{i,k}A^f_{i,k})^\theta}{(\alpha_i)^\theta + \sum_{k=1}^K (p_{i,k}A^f_{i,k})^\theta}\right]^{(\theta-1)/\theta}.$$

Summing over fields gives country-crop output $Q_{i,k}$, and market clearing requires $Q_{i,k} = \sum_j d_{ij,k}C_{ij,k}$ for every (i,k).

Intuition: Why this is comparative advantage at scale

The land-share formula is a many-crop analogue of the DFS/EK cutoff rule: a plot grows whichever crop offers it the highest *relative* return $p_{i,k}A^f_{i,k}$, and θ governs how decisively plots sort by that comparative advantage. A country's aggregate crop mix, and hence its exports, is the sum of millions of such micro-decisions — which is why 1.7 million field-level Fréchet draws, rather than a single national productivity number per crop, are needed to take the model to data.

Understanding Quiz 1 — Lesson 1

Q1.1 FAO-GAEZ "potential yield" $A^f_{i,k}$ is best described as:
- (A) the yield actually observed on a field
- (B) the mean yield if the entire field were devoted to crop k, an agro-climatic (not socio-economic) object
- (C) the labor cost of producing crop k
- (D) a market price

Q1.2 The two margins of climate adaptation in CDS are:
- (A) saving and investment
- (B) production (switching crops within a country) and trade (switching source countries)

(C) migration and fertility
(D) monetary and fiscal policy

Q1.3 A plot is allocated to crop k in equilibrium only when:
(A) its productivity draw for k delivers the highest return among all crops and the outside option
(B) the government mandates it
(C) $\theta = 0$
(D) all crops are grown with equal probability

Q1.4 Because plots select into crops based on favorable draws, conditional mean productivity on planted land is:
(A) equal to the unconditional mean $A^f_{i,k}$
(B) lower than the unconditional mean $A^f_{i,k}$
(C) higher than the unconditional mean $A^f_{i,k}$, scaling with $(\pi^f_{i,k})^{-1/\theta}$
(D) undefined

2 Lesson 2: Estimation and the Welfare Consequences of Climate Change

2.1 Identifying demand elasticities

Derivation: Two nested gravity regressions

Taking logs of the expenditure-share equations delivers two nested gravity specifications:

$$\ln(X_{ij,k}/X_{j,k}) = M_{j,k} + (1-\sigma)\ln p_{i,k} + \varepsilon_{ij,k}, \qquad \ln(X_{j,k}/X_j) = M_j + (1-\kappa)\ln P_{j,k} + \varepsilon_{j,k}.$$

OLS is biased because $p_{i,k}$ and $P_{j,k}$ are equilibrium prices correlated with the demand shocks ε. CDS instruments with **average potential yield** of crop k in country i: an agro-climatic supply shifter plausibly uncorrelated with demand shocks. This is the identical logic to the price-based route to θ in the Ricardo-to-EK companion (Lesson 12) — find a variable that shifts the regressor without shifting the error.

2.2 Identifying the supply (Fréchet dispersion) parameter

Derivation: Matching land allocation and output by NLS

Because $A^f_{i,k}$ is directly observed for every field (unlike unobserved labor productivity draws in EK), only α_i and θ remain to be estimated. First, for any candidate θ, choose $\alpha_i(\theta)$ so that model-predicted agricultural land exactly matches observed agricultural land,

$$L^{agr}_i(\theta) \equiv \sum_{k,f} L^f_i \frac{(p_{i,k}A^f_{i,k})^\theta}{(\alpha_i(\theta))^\theta + \sum_k (p_{i,k}A^f_{i,k})^\theta} = L^{agr}_i \text{ (data)}.$$

Then choose θ by nonlinear least squares to match log output by country-crop:

$$\min_{\theta} \sum_{i,k} \left[\ln Q_{i,k}(\theta) - \ln Q_{i,k}\right]^2 \quad \text{s.t. } L_i^{agr}(\theta) = L_i^{agr} \ \forall i.$$

Empirical Fact: Estimated parameters (50 countries, 10 crops, 71% of world output)

$\sigma = 5.40$ (across-country substitution within a crop), $\kappa = 2.82$ (across-crop substitution), $\theta = 2.46$ (land productivity dispersion). A low θ relative to trade elasticities in manufacturing (recall $\theta \approx 4$–8 in EK) signals *substantial* heterogeneity in land productivity across crops within a field — strong scope for comparative-advantage-driven reallocation.

2.3 Counterfactual climate change

Derivation: Three counterfactual scenarios

Feed FAO-GAEZ's 2071–2100 productivity changes $\{A_{i,k}^{f\prime}\}$ into the estimated model and re-solve for equilibrium prices, comparing three cases:

1. **Full adjustment**: both $\pi_{i,k}^{f}$ (crop mix) and trade shares respond optimally.
2. **No production adjustment**: land shares $\pi_{i,k}^{f}$ are frozen at baseline values; only trade reallocates.
3. **No trade adjustment**: bilateral trade shares are frozen; only domestic crop-switching reallocates.

Empirical Fact: Headline welfare results

World real food consumption falls by **0.26%** under full adjustment. The loss is roughly **three times as large** if farmers cannot reallocate production across crops within fields (no production adjustment), showing that within-country crop switching is a first-order adaptation margin. Effects are highly heterogeneous across countries: some (e.g. Canada, Russia) gain, while others (e.g. Malawi, at -49%) face catastrophic losses.

Understanding Quiz 2 — Lesson 2

Q2.1 CDS instrument prices with average potential yield because it:
(A) shifts demand directly
(B) is an agro-climatic supply shifter plausibly excluded from the demand error
(C) is set by governments
(D) equals the wage

Q2.2 Because $A_{i,k}^{f}$ is observed directly from FAO-GAEZ, estimating the supply side reduces to recovering:
(A) only θ
(B) only α_i
(C) (α_i, θ) jointly, matching agricultural land and output

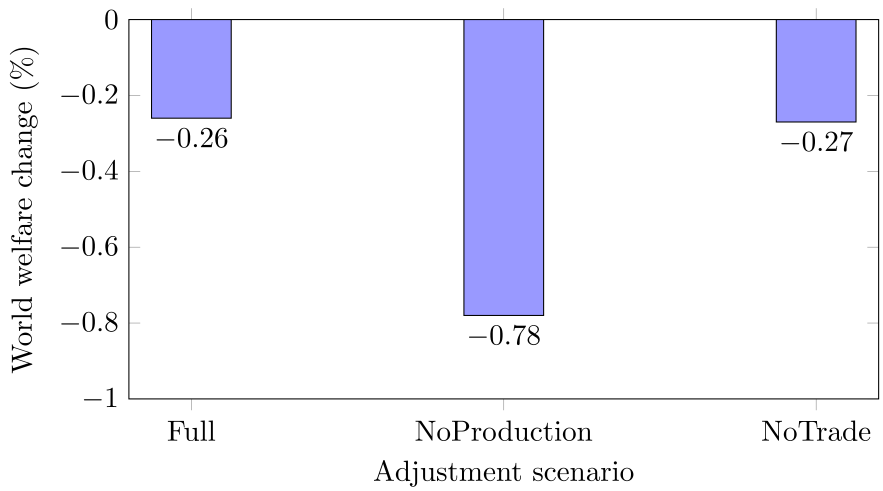

Figure 2: World welfare change under climate change (schematic, based on CDS Table 3 world totals): freezing production adjustment triples the loss, while freezing trade adjustment barely changes it — in this calibration the *production* margin (crop switching) matters far more than the *trade* margin at the world level, even though both matter enormously for individual countries.

(D) nothing — the supply side is fully known

Q2.3 Freezing production adjustment (no crop switching) in the climate counterfactual:
(A) has no effect on welfare
(B) roughly triples the world welfare loss relative to full adjustment
(C) eliminates the welfare loss
(D) only affects trade shares

Q2.4 The estimated $\theta = 2.46$ for land, compared to typical EK estimates of $\theta \approx 4$–8 for manufacturing, implies:
(A) land productivity is less dispersed than manufacturing productivity
(B) land productivity is *more* dispersed across crops within a field, giving strong scope for comparative-advantage reallocation
(C) θ is not comparable across sectors
(D) agriculture has no comparative advantage

Part II
Trade, Technology Adoption, and Agricultural Productivity

This part is based on Farrokhi and Pellegrina (2023, *JPE*), "Trade, Technology, and Agricultural Productivity." While Part I asked how trade in agricultural *outputs* lets countries specialize according to comparative advantage, this part asks a complementary question: how does trade in agricultural *inputs* (fertilizer, machinery, chemicals) let countries adopt higher-yielding *technologies*? The paper nests a discrete technology choice inside the same Fréchet land-allocation framework used in Part I.

3 Lesson 3: Four Empirical Patterns and the Technology-Adoption Model

3.1 Four motivating patterns

Empirical Fact: Pattern 1 — Input intensity rises with income

The cost share of agricultural inputs and input-per-land (or per-labor) use both rise with GDP per capita, while labor-per-land falls with GDP per capita (slopes of $+0.09$, $+1.48$, and -0.80 respectively across a 65-country cross-section).

Empirical Fact: Pattern 2 — Heavy reliance on imported inputs

On average **two-thirds** of spending on farm machinery, fertilizer, and pesticides is imported; imports average 65% of expenditure, while exports of these inputs are concentrated in a small number of countries (top 10 countries supply 77–85% of world exports by input category).

Empirical Fact: Pattern 3 — Large, field-heterogeneous yield premia

Modern-technology potential yields exceed traditional-technology potential yields by a factor of 4–7 depending on the crop (e.g. $4.0\times$ for potato, $6.7\times$ for soy), and this premium is highly dispersed *across fields even within the same country* — it is not simply a function of national income.

Empirical Fact: Pattern 4 — Adoption falls with distance to trade hubs

Regressing modern-technology land share on log trade cost to the nearest hub, with country-crop fixed effects and potential-yield controls, gives a coefficient of -1.312 (s.e. 0.180): remoteness from ports and cities predicts lower adoption of modern technology, holding fixed the crop and its agro-climatic potential.

Intuition: Reading the four facts together

Pattern 1 says technology differs systematically with development; Pattern 2 says this is not autarkic — it depends on importing inputs; Pattern 3 says the *potential* for modernization is everywhere similar in scale, so income differences in adoption must reflect economic incentives (input prices, market access) rather than agronomy; Pattern 4 confirms market access (domestic and international) is one such incentive. Together they motivate a model where technology choice is an equilibrium outcome of factor and input prices, not an exogenous productivity parameter.

3.2 Environment: crop *and* technology choice on every plot

Definition: Setup

Countries $i \in N$, endowed with land L_n, labor N_n, and raw fertilizer V_n. Goods span a non-agricultural good ($g = 0$), agricultural inputs $j \in \mathcal{J}$ (fertilizer, machinery, pesticide), and agricultural outputs $k \in \mathcal{K}$. Consumption nests three CES tiers: agriculture vs. non-agriculture, crop vs. crop, and origin country vs. origin country (as in Part I). Production of crop k with technology $\tau \in \{0, 1\}$ (traditional, modern) on plot ω uses land, labor, and agricultural inputs via

$$Q^f_{i,k\tau}(\omega) = \bar{q}_{k\tau}\left(z^f_{i,k\tau}(\omega)\right)\left(L^f_{i,k\tau}(\omega)\right)^{\gamma^L_{k\tau}}\left(N^f_{i,k\tau}(\omega)\right)^{\gamma^N_{k\tau}}\prod_{j\in\mathcal{J}}\left(M^{j,f}_{i,k\tau}(\omega)\right)^{\gamma^M_{k\tau}\lambda^j_k},$$

with $\gamma^M_{k0} = 0$ (traditional farming uses no purchased inputs) and $\gamma^M_{k1} > 0$ (modern farming does), and $\gamma^L_{k\tau} + \gamma^N_{k\tau} + \gamma^M_{k\tau} = 1$.

Derivation: Returns and the generalized Fréchet

Cost minimization plus zero profits pin down plot-level "returns" to each crop-technology pair, $r^f_{i,k\tau}(\omega) = z^f_{i,k\tau}(\omega)\, h_{i,k\tau}$, where $h_{i,k\tau} = p_{i,k}\left(w_i/p_{i,k}\right)^{-\gamma^N_{k\tau}/\gamma^L_{k\tau}}\left(m_{i,k}/p_{i,k}\right)^{-\gamma^M_{k\tau}/\gamma^L_{k\tau}}$ bundles the output price and input-price-adjusted composite cost. Productivity draws $z^f_i(\omega) = [z^f_{i,k\tau}(\omega)]_{k,\tau}$ follow a **nested (generalized) Fréchet** distribution with an *upper nest* across crops (dispersion θ_1) and a *lower nest* across technologies within a crop (dispersion θ_2):

$$\Pr\left(z^f_i(\omega) \le z^f_i\right) = \exp\left(-\bar{\phi}\sum_k \Gamma_k(z^f_{i,k})^{-\theta_1}\right), \qquad \Gamma_k(z^f_{i,k}) = \left[\sum_\tau \left(\frac{z^f_{i,k\tau}}{a^f_{i,k\tau}}\right)^{-\theta_2}\right]^{-1/\theta_2}.$$

Intuition: What θ_1 and θ_2 each control

θ_1 governs dispersion *across crops* within a field: a higher θ_1 means crops are more agronomically alike within a field, so producers respond less to relative-return changes by switching crops (this is the same θ as in CDS's single-nest model). θ_2 governs dispersion *across technologies within a given crop*: a higher θ_2 means traditional and modern draws for the same crop are more correlated, so producers respond more sharply to changes in the *relative* return between adopting and not adopting modern technology — an extensive-margin technology-adoption elasticity, analogous to θ in EK but for technology rather than trading partners.

Derivation: Closed-form land shares (two-stage nested logit/Fréchet)

$$\pi^f_{i,k\tau} = \underbrace{\frac{(H^f_{i,k})^{\theta_1}}{\sum_k (H^f_{i,k})^{\theta_1}}}_{\alpha^f_{i,k}:\ \text{share in crop } k} \times \underbrace{\frac{(a^f_{i,k\tau}h_{i,k\tau})^{\theta_2}}{(H^f_{i,k})^{\theta_2}}}_{\alpha^f_{i,k\tau}:\ \text{share in tech } \tau \text{ given crop } k}, \qquad H^f_{i,k} \equiv \left[\sum_\tau (a^f_{i,k\tau}h_{i,k\tau})^{\theta_2}\right]^{1/\theta_2}.$$

Summing plot-level output over technologies and fields, as in Part I, delivers country-crop supply $Q_{i,k} = \sum_{f,\tau} Q^f_{i,k\tau}$, and a corresponding **production possibility frontier** whose curvature is governed by θ_1 across crops and by θ_2 across technologies within a crop.

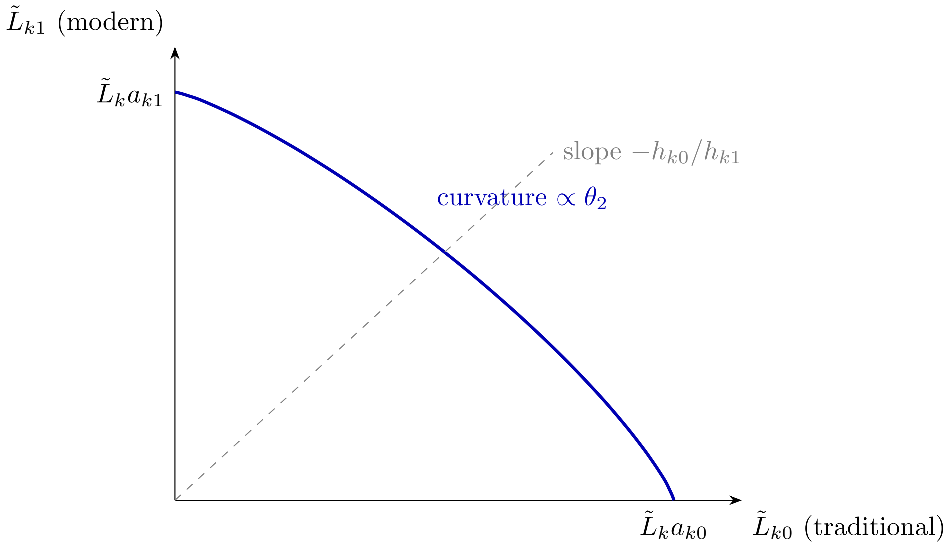

Figure 3: The lower-nest PPF between traditional and modern technology within crop k: curvature is governed by θ_2, exactly analogous to the two-technology production-possibility curve of a nested-CES/Fréchet discrete-choice model.

Understanding Quiz 3 — Lesson 3

Q3.1 Empirical Pattern 3 (large yield premia that don't vary systematically with GDP per capita) is important because it shows:
(A) poor countries have no agronomic potential for modern technology
(B) the *potential* for modernization is broadly similar everywhere, so cross-country adoption gaps must reflect economic incentives, not agronomy
(C) rich countries have higher potential yields
(D) technology adoption is random

Q3.2 In the generalized Fréchet, θ_1 and θ_2 respectively govern dispersion:
(A) across countries; across crops
(B) across crops within a field; across technologies within a crop
(C) across technologies; across crops
(D) across fields; across countries

Q3.3 Traditional technology differs from modern technology in the production function because:
(A) traditional technology uses more land
(B) $\gamma^M_{k0} = 0$: traditional farming uses no purchased agricultural inputs, while modern farming does ($\gamma^M_{k1} > 0$)
(C) modern technology uses no labor
(D) they have identical input shares

Q3.4 Pattern 4 (adoption falls with distance to trade hubs) is evidence for:
(A) the role of domestic and international market access in technology adoption, holding potential yield fixed
(B) random measurement error
(C) that only rich countries have hubs
(D) that technology does not depend on geography

4 Lesson 4: Gains from Trade, Estimation, and Globalization Counterfactuals

4.1 A clean decomposition of the gains from trade

Derivation: Trade gains vs. technology gains

In a pared-down one-good, one-field version of the model (autarky permits only traditional technology, since agricultural inputs must be imported), the total gains from trade decompose multiplicatively into an Arkolakis-Costinot-Rodríguez-Clare (ACR) trade term and a Fréchet technology-adoption term:

$$\boxed{G_i = 1 - \underbrace{(\lambda_{ii})^{\frac{1}{\sigma-1}}}_{\text{Trade (ACR)}} \underbrace{(\alpha_{i,0})^{\frac{1}{\theta_2}}}_{\text{Technology (FP)}}}$$

where λ_{ii} is the domestic expenditure share and $\alpha_{i,0}$ is the share of land under traditional technology. Both terms are *sufficient statistics*: everything about foreign productivities, trade costs, and input prices compresses into these two observable shares.

Intuition: Why the gains estimate roughly quintuples for Colombia

Evaluated at Colombian data ($\lambda_{ii} = 0.85$, $\alpha_{i,0} = 0.55$) and estimated elasticities ($\sigma = 5.7$, $\theta_2 = 4.5$), the ACR-only formula gives gains of just 3.4% of income, while adding the technology term raises the estimate to 18.6%. The classical ACR formula, built for trade in *final* goods, misses gains that flow through trade in *inputs* enabling technology upgrading — an entirely separate channel that the data show is roughly as large as the classical channel.

4.2 Taking the model to data

Derivation: Two informative data moments for (θ_1, θ_2)

The paper estimates $\Theta = (\theta_1, \theta_2)$ by minimum distance, matching a set of moments $m(\Theta)$ to their data counterparts, subject to calibration constraints $c(\Theta) = 0$ (e.g. using FAO-GAEZ potential yields as observed productivity shifters $a^f_{i,k\tau}$). Two moments are particularly informative:

- **Land share across crops vs. relative potential yield**: regressing log relative land share on log relative potential yield gives a slope close to **1.00** (traditional) and **0.73** (modern), pinning down θ_1.

- **Modern land share vs. log internal trade cost**: a slope of -10.67 (95% CI $[-13.47, -7.87]$) is highly informative about θ_2, since internal trade costs shift the relative price of the imported input bundle.

Estimated values: $\theta_1 = 1.38$ (s.e. 0.19), $\theta_2 = 4.51$ (s.e. 0.45); demand-side estimates $\sigma_k = 5.76$, $\kappa = 4.16$.

4.3 Counterfactual: reversing globalization in agricultural inputs and outputs

Empirical Fact: Aggregate effects of rolling trade costs back to 1980 levels

Rolling agricultural trade costs (both inputs and outputs) back to their 1980 levels would: raise the domestic expenditure share on agricultural inputs by 18.8 points and on outputs by 6.9 points; lower the share of land in modern technology by 5.1 points and average yields by 8.5%; and lower *global welfare* by 3.3%. Decomposing further: reversing *only input* trade costs delivers a welfare loss of 1.6%, and reversing *only output* trade costs also delivers 1.6% — the two channels are of comparable, and largely additive, importance.

Intuition: Two different globalizations, two different beneficiaries

Globalization in agricultural *outputs* operates through international crop specialization and benefits low-income countries most (they spend a larger expenditure share on food, so cheaper imported food delivers large welfare gains). Globalization in agricultural *inputs* operates through technology adoption and benefits *middle*-income countries most: high-income countries are already near- universally modernized (little room to adopt further), while the poorest countries have so little modern-technology land that raising productivity *on* that land moves little of the aggregate. It is precisely middle-income countries with intermediate modern-land shares who gain most from cheaper inputs.

Understanding Quiz 4 — Lesson 4

Q4.1 In the gains-from-trade decomposition $G_i = 1 - \lambda_{ii}^{1/(\sigma-1)} \alpha_{i,0}^{1/\theta_2}$, the second term captures gains from:
(A) lower trade costs on final goods only
(B) access to imported agricultural inputs, enabling technology adoption
(C) labor mobility
(D) government subsidies

Q4.2 The moment "modern land share vs. log internal trade cost" (slope ≈ -10.7) is most informative about:
(A) σ
(B) κ
(C) θ_1
(D) θ_2

Q4.3 Reversing globalization in agricultural inputs *only* (not outputs) back to 1980 levels:
(A) raises global welfare
(B) lowers modern-technology land share and yields, and lowers welfare by about 1.6%

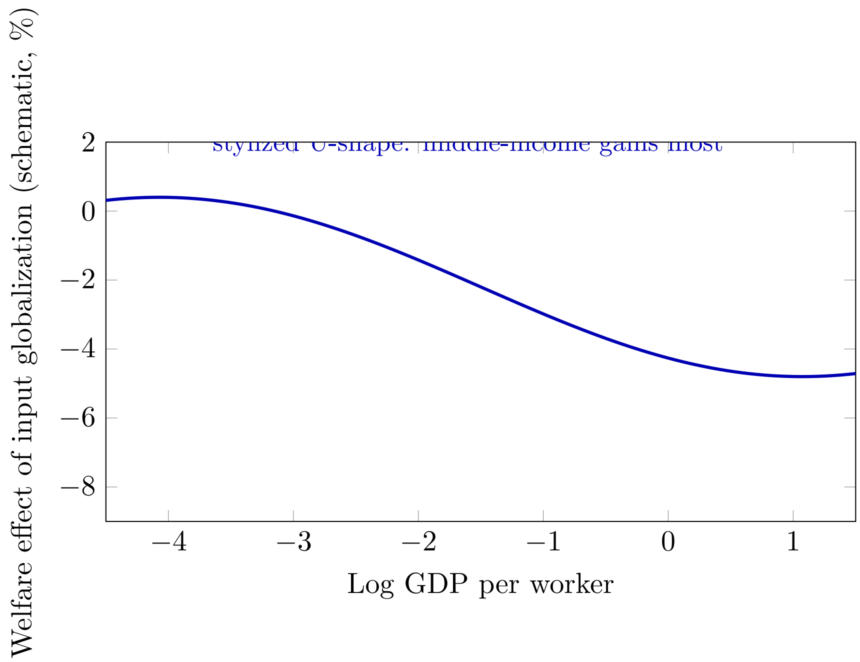

Figure 4: Schematic rendering of Farrokhi–Pellegrina's cross-country welfare pattern for the "input-only" globalization counterfactual (their Figure, "Cross-country Distributional Effects"): a roughly U-shaped (or hump-shaped in gains) relationship with income, in contrast to the monotonically declining pattern for output-only globalization.

(C) has no effect on technology adoption
(D) only affects rich countries

Q4.4 Output-side globalization benefits _____ most, while input-side globalization benefits _____ most:
(A) rich countries; poor countries
(B) poor countries (high food expenditure share); middle-income countries (intermediate room to adopt technology)
(C) all countries equally in both cases
(D) middle-income countries; poor countries

Part III

Structural Transformation over the Growth Path

This part is based on Herrendorf, Rogerson, and Valentinyi's *Handbook of Economic Growth* chapter on structural transformation. Where Parts I–II held aggregate sectoral composition fixed and asked how trade reshapes it, this part builds the closed-economy benchmark that explains *why* economies move from agriculture to manufacturing to services as they grow in the first place — the backdrop against which the climate/trade forces of the other parts operate.

5 Lesson 5: Stylized Facts and the Benchmark Multi-Sector Growth Model

5.1 What is being explained, and how is it measured

Definition: Structural transformation

Structural transformation is the reallocation of economic activity across agriculture, manufacturing, and services that accompanies modern economic growth. It is measured three ways — **employment shares**, **value-added shares** (nominal or real), and **final consumption expenditure shares** — which are conceptually distinct (the first two are production-side; the third is consumption-side) and can diverge once trade or input-output linkages are present.

Empirical Fact: The long-run and cross-country patterns

Across both long time series and recent cross-country data: agriculture's share declines monotonically with development; services' share rises monotonically; manufacturing is **hump-shaped**, rising then falling. The pattern appears in both employment and nominal value added, though real and nominal value-added shares can diverge once relative prices move.

5.2 The benchmark model: separating growth from reallocation

Definition: Environment

A representative household solves $\max \sum_t \beta^t \log C_t$ subject to $P_t C_t + K_{t+1} = (1-\delta+R_t)K_t + W_t$. There are four sectors: agriculture a (necessities, strong income effects), manufacturing m, services s (luxuries), and investment x (the engine of aggregate growth). Composite consumption is non-homothetic CES,

$$C_t = \left[\omega_a^{1/\varepsilon}(c_{at} - \bar{c}_a)^{\frac{\varepsilon-1}{\varepsilon}} + \omega_m^{1/\varepsilon} c_{mt}^{\frac{\varepsilon-1}{\varepsilon}} + \omega_s^{1/\varepsilon}(c_{st} + \bar{c}_s)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\bar{c}_a > 0$ is a subsistence requirement (food is a necessity) and $\bar{c}_s > 0$ a services "entry cost" (services are a luxury); ε governs substitution. Each sector is Cobb-Douglas, $c_{it} = k_{it}^{\theta}(A_{it} n_{it})^{1-\theta}$, and with free factor mobility capital-labor ratios equalize across sectors: $k_{it}/n_{it} = K_t$ for all i.

Derivation: Aggregation: the economy behaves like a one-sector model

Despite four sectors, equalized capital-labor ratios imply the economy aggregates: $Y_t = X_t + P_t C_t = K_t^{\theta} A_{xt}^{1-\theta}$, with $R_t = \theta K_t^{\theta-1} A_{xt}^{1-\theta}$ and $W_t = (1-\theta) K_t^{\theta} A_{xt}^{1-\theta}$. The Euler equation $\frac{1}{\beta}\frac{P_t C_t}{P_{t-1} C_{t-1}} = 1 - \delta + R_t$ governs the **intertemporal** problem (consumption vs. saving), driven entirely by investment-sector productivity A_{xt}. A separate **static** problem allocates $P_t C_t$ across c_a, c_m, c_s each period. **Key insight: growth $\neq$ structural transformation** — the same K_t path is consistent with many different sectoral compositions.

Derivation: Relative prices are pinned down by technology

Sectoral productivity growth $A_{i,t+1}/A_{it} = 1 + \gamma_i$ translates directly into relative prices:

$$p_{it} = \left(\frac{A_{xt}}{A_{it}}\right)^{1-\theta}, \qquad i = a, m, s.$$

A **generalized balanced growth path (GBGP)** requires only that the real interest rate R_t be constant — a weaker requirement than exact balanced growth, which would demand constant sectoral shares. Under GBGP, aggregate variables grow at a constant rate γ_x while sectoral shares evolve freely.

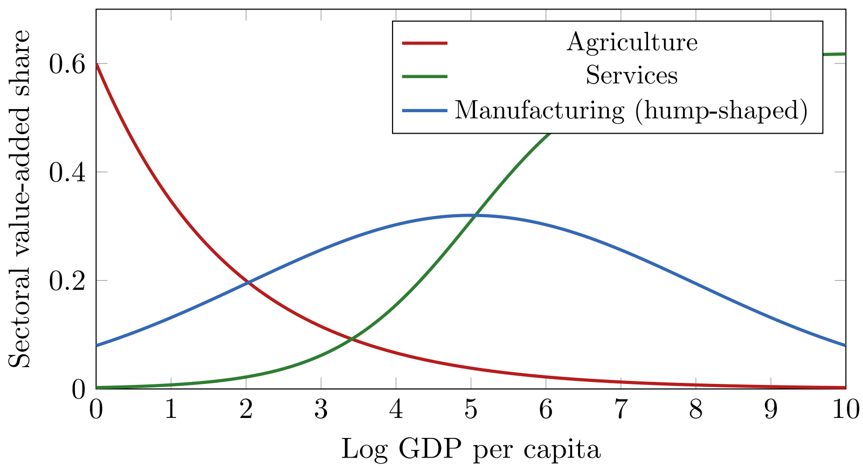

Figure 5: Stylized version of Herrendorf-Rogerson-Valentinyi's long-run and cross-country pattern: agriculture declines monotonically, services rise monotonically, and manufacturing is hump-shaped in the level of development.

Understanding Quiz 5 — Lesson 5

Q5.1 The three ways of measuring structural transformation (employment, value added, expenditure shares) are:
- (A) always identical
- (B) conceptually distinct — the first two are production-side, the third consumption-side, and they can diverge with trade or IO linkages
- (C) only relevant in autarky
- (D) interchangeable if $\varepsilon = 1$

Q5.2 Manufacturing's share of value added over the growth path is typically:
- (A) monotonically rising
- (B) monotonically falling
- (C) hump-shaped
- (D) constant

Q5.3 The key decomposition "growth $\neq$ structural transformation" means:
- (A) growth is driven by the intertemporal (saving) problem via A_{xt}, while transformation comes from the static allocation problem across c_a, c_m, c_s

(B) structural transformation causes all aggregate growth
(C) they are the same object
(D) neither is affected by technology

Q5.4 A generalized balanced growth path (GBGP) requires:
(A) constant sectoral employment shares
(B) a constant real interest rate, while sectoral shares are allowed to evolve
(C) zero productivity growth
(D) constant relative prices

6 Lesson 6: Income vs. Price Mechanisms, and the Role of International Trade

6.1 Two classic mechanisms

Derivation: Case 1: pure income effects (Kongsamut-Rey-Xie)

If productivity growth is common across consumption sectors ($\gamma_a = \gamma_m = \gamma_s$) and $\varepsilon = 1$, relative prices are constant but sectoral shares still move because preferences are non-homothetic (Stone-Geary):

$$c_{at} = \omega_a \frac{P_t C_t}{p_{at}} + \bar{c}_a, \quad c_{mt} = \omega_m \frac{P_t C_t}{p_{mt}}, \quad c_{st} = \omega_s \frac{P_t C_t}{p_{st}} - \bar{c}_s.$$

This explains declining agriculture and rising services well but struggles to generate a manufacturing hump on its own.

Derivation: Case 2: pure relative-price effects (Ngai-Pissarides)

If $\bar{c}_a = \bar{c}_s = 0$ (homothetic), transformation comes entirely from differential TFP growth, $\gamma_a > \gamma_m > \gamma_s$ (agriculture's TFP grows fastest, services' slowest). With $\varepsilon < 1$ (low substitutability), the sector with the fastest productivity growth *shrinks* in value-added share even as its output rises, because prices fall faster than quantities rise. This mechanism naturally explains why nominal and real value-added shares diverge.

Intuition: The empirical verdict

Sectoral TFP growth differences are empirically real (agriculture typically has the fastest TFP growth post-1970, services the slowest) — evidence favoring price effects. But capital shares, substitution patterns, and income effects also matter. The chapter's methodological message is that no single mechanism is universally correct; what matters is that theory be mapped carefully to the correct empirical object (employment vs. value added vs. expenditure).

6.2 Opening the economy: how trade reshapes structural transformation

Derivation: Breaking the link between production and expenditure shares

In the closed-economy benchmark, domestic production must equal domestic absorption in every sector, so production and expenditure shares move together by construction. Once trade is allowed, firms specialize by comparative advantage while households consume according to preferences, so

$$\text{production shares} \neq \text{consumption expenditure shares.}$$

A country with strong productivity growth in manufacturing can see a *rising* manufacturing value-added or employment share (driven by specialization and exports) even while its long-run consumption bundle continues to shift toward services — this is one explanation for why some countries display a pronounced manufacturing hump while others do not.

Intuition: Openness as an independent driver

Because sectoral labor allocation under trade depends on relative productivity growth, specialization pattern, and access to world markets — not domestic demand alone — openness does not just change trade volumes; it can change the entire *path* of structural transformation, even holding household preferences fixed. This is the bridge connecting Part IV (Herrendorf et al.) to Parts I–II (Costinot et al.; Farrokhi-Pellegrina): trade reallocates *production* across countries according to comparative advantage, and that reallocation feeds back into each country's observed sectoral trajectory.

Empirical Fact: Other extensions

The chapter also surveys: **labor mobility frictions** (costs of moving across sectors can slow reallocation even when relative returns favor it — directly relevant to the Roy-Fréchet selection models of Parts I–II and V–VI); **goods mobility and transport costs** (lower transport costs accelerate labor movement out of agriculture); and applications to regional convergence, aggregate productivity, wage inequality, and business cycles.

Understanding Quiz 6 — Lesson 6

Q6.1 The Kongsamut-Rey-Xie mechanism relies mainly on:
(A) differential TFP growth across sectors
(B) non-homothetic (Stone-Geary) preferences with roughly common TFP growth
(C) trade liberalization
(D) labor mobility frictions

Q6.2 The Ngai-Pissarides mechanism explains diverging real vs. nominal value-added shares through:
(A) income effects only
(B) differential TFP growth combined with low substitutability ($\varepsilon < 1$) across sectors
(C) government policy
(D) constant relative prices

Q6.3 Opening the economy to trade breaks the tight link between production and expenditure shares because:

(A) firms now specialize by comparative advantage while consumption follows preferences, so the two need not coincide
(B) trade eliminates all sectors but one
(C) prices become fixed
(D) closed-economy identities still hold exactly

Q6.4 A country with strong manufacturing TFP growth that opens to trade may:

(A) never show a manufacturing hump
(B) show a rising manufacturing share via specialization/exports even as its consumption continues shifting toward services
(C) automatically deindustrialize
(D) have identical experience to a closed economy

Part IV
Selection and Cross-Country Agricultural Productivity Gaps

This part is based on Lagakos and Waugh (2013, *AER*), "Selection, Agriculture, and Cross-Country Productivity Differences." It provides the closed-economy **worker-selection** counterpart to the field-level **land-selection** models of Parts I–II: instead of asking which *crop* a plot of land grows, it asks which *sector* a worker chooses, and shows that this choice alone can make measured agricultural productivity gaps look far larger than the underlying efficiency gaps that Parts I–III take as exogenous inputs.

7 Lesson 7: The Puzzle and the Roy Model of Sectoral Selection

7.1 The central empirical fact

Empirical Fact: A 10.7× amplification

Comparing 90th- to 10th-percentile countries: the agricultural labor-productivity gap is about **45×**, versus only **4×** in non-agriculture (aggregate gap: 22×). Agriculture's employment share falls from 78% in poor countries to 3% in rich countries. Cross-country productivity differences are thus more than **ten times larger** in agriculture than in non-agriculture — the central puzzle the paper resolves through worker selection.

Intuition: The core idea in one line

Poor countries have large agricultural employment shares, so many agricultural workers there are *not* especially productive at farming (subsistence forces them in regardless). Rich countries have tiny agricultural employment shares, so only workers with a strong *comparative* advantage in farming remain. Measured agricultural productivity is therefore mechanically dragged down in poor countries and pushed up in rich countries by **who selects into the sector** — even if the underlying, economy-wide efficiency gap between rich and poor is much smaller.

7.2 The model: subsistence, heterogeneous workers, Roy selection

Definition: Ingredients

Two sectors, agriculture a and non-agriculture n. Preferences feature a subsistence food requirement: $U_i = \log(c_i^a - \bar{a}) + \nu \log(c_i^n)$. Workers are heterogeneous in a pair of sector-specific efficiencies $(z_i^a, z_i^n) \sim G(z^a, z^n)$ and choose whichever sector pays more (Roy selection). Sectoral output is linear in effective labor, $Y_a = AL_a$, $Y_n = AL_n$, with $L_a = \int_{i \in \Omega_a} z_i^a \, dG_i$ and similarly for n, so *measured* labor productivity $Y_a/N_a = A \cdot L_a/N_a$ depends on both a common efficiency level A and *which workers* select into agriculture.

Derivation: The selection rule

With wages per efficiency unit $w_a = p_a A$, $w_n = A$, a worker chooses non-agriculture iff $z_i^n A \geq z_i^a p_a A$, i.e.

$$\boxed{\frac{z_i^n}{z_i^a} \geq p_a}.$$

Workers sort purely on comparative advantage; the relative price of agriculture p_a is the threshold that clears the labor market.

Derivation: Proposition 1: the relative price of agriculture

For rich country R and poor country P with $A_R > A_P$: $\boxed{p_a^P > p_a^R}$. Because poor countries have lower income, subsistence food demand looms larger, so agriculture's relative price must be higher to draw enough workers into food production.

Understanding Quiz 7 — Lesson 7

Q7.1 The central empirical puzzle Lagakos-Waugh resolve is:
- (A) why aggregate productivity gaps are zero
- (B) why cross-country agricultural productivity gaps (≈45×) are far larger than non-agricultural gaps (≈4×)
- (C) why rich countries have more farmers
- (D) why subsistence consumption does not exist

Q7.2 A worker chooses non-agriculture in the Roy model when:
- (A) $z_i^n/z_i^a \geq p_a$
- (B) $z_i^n/z_i^a \leq p_a$
- (C) $z_i^a > z_i^n$ always
- (D) wages are equal in both sectors regardless of z

Q7.3 Proposition 1 states that poor countries have a _____ relative price of agriculture than rich countries:
- (A) lower
- (B) higher, because subsistence food demand is more binding

(C) identical
(D) undefined

Q7.4 Measured agricultural labor productivity depends on:
(A) only the common efficiency parameter A
(B) both the common efficiency A and the selection of which workers choose agriculture
(C) only the subsistence parameter $\bar{a}$
(D) nothing — it is exogenous

8 Lesson 8: Amplification, Fréchet Quantification, and Results

8.1 Proposition 2: selection amplifies the agricultural gap

Derivation: Amplification under positive sorting

If comparative and absolute advantage are positively correlated (higher z^a/z^n types also have higher z^a on average), then

$$\boxed{\frac{Y_a^R/N_a^R}{Y_a^P/N_a^P} > \frac{A_R}{A_P}} \qquad \text{and} \qquad \boxed{\frac{Y_n^R/N_n^R}{Y_n^P/N_n^P} < \frac{A_R}{A_P}}.$$

Selection makes the agricultural productivity gap *larger* than the underlying efficiency gap A_R/A_P, while shrinking the non-agricultural gap below it.

8.2 A Fréchet analytical example

Derivation: Closed-form employment shares and selection

With $G(z^a) = e^{-(z^a)^{-\theta}}$, $G(z^n) = e^{-(z^n)^{-\theta}}$, the agricultural employment share is $\pi_a = 1/(p_a^{-\theta} + 1)$, giving the log-linear relation $\log(\pi_a/\pi_n) = \theta \log(p_a)$. Expected productivity conditional on selecting into agriculture is $\mathbb{E}(z^a \mid a) = \gamma\, \pi_a^{-1/\theta}$: as $\pi_a \downarrow$ (rich countries), selected agricultural productivity rises mechanically — exactly the same Fréchet-selection logic used for land in Parts I–II, now applied to labor.

Empirical Fact: Quantitative results: data vs. model

Calibrated to match the 90-10 gaps, the model produces an agricultural/non-agricultural productivity-gap *ratio* of **2.2** (model: 29 vs. 13) against a data target of **10.7** (45 vs. 4); a model with no selection (equal composition) delivers a ratio of exactly **1.0** (both 19). Selection alone therefore explains roughly *half* of the observed amplification, and decomposing further: rich-country agricultural workers have average agricultural productivity about **1.55**$\times$ the population mean, versus **1.00**$\times$ for poor-country agricultural workers — the mirror image in non-agriculture is **1.01**$\times$ (rich) vs. **1.42**$\times$ (poor).

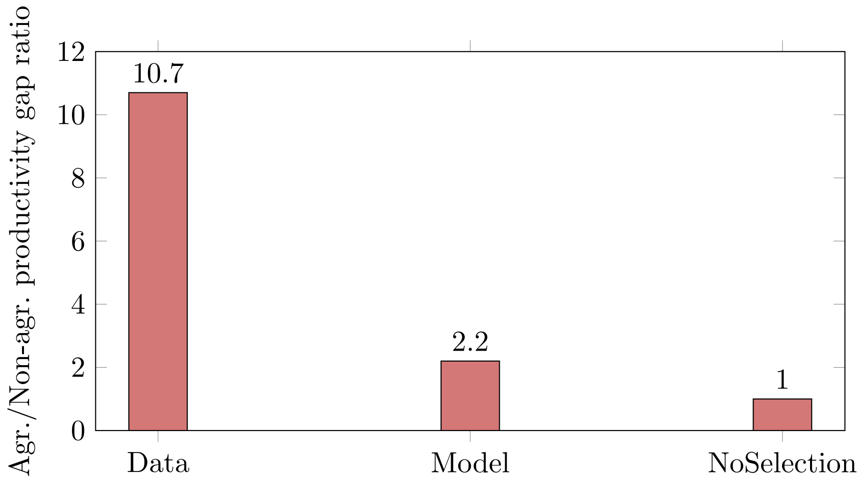

Figure 6: Lagakos-Waugh's benchmark quantitative result: worker selection alone generates roughly half of the observed 10.7× amplification of agricultural productivity gaps relative to non-agriculture, even though it cannot fully close the gap between model and data.

Intuition: Connecting back to trade and structural transformation

Lagakos-Waugh's Fréchet selection machinery is mathematically the same Roy-Fréchet apparatus that Costinot-Donaldson-Smith apply to land and Farrokhi-Pellegrina apply to crop-technology pairs. The economics differ (labor vs. land selection) but the lesson is the same: comparing measured productivity across units that are *not* randomly assigned to a sector or crop can badly overstate the underlying technology gap, because composition itself responds to the very forces (income, subsistence, prices) being studied. This is also the mechanism underlying the "food problem" in Part VI (Nath 2025): poor countries keep low-comparative-advantage workers in agriculture precisely because they cannot escape it.

Understanding Quiz 8 — Lesson 8

Q8.1 Proposition 2 says selection makes the agricultural productivity gap:
(A) exactly equal to A_R/A_P
(B) larger than A_R/A_P, while the non-agricultural gap becomes smaller than A_R/A_P
(C) unrelated to A_R/A_P
(D) always equal to one

Q8.2 In the Fréchet example, $\mathbb{E}(z^a \mid a) = \gamma\pi_a^{-1/\theta}$ implies that when a country's agricultural employment share π_a falls:
(A) selected agricultural productivity falls
(B) selected agricultural productivity mechanically rises (fewer, more selected farmers)
(C) nothing changes
(D) θ must be zero

Q8.3 The calibrated model produces an agriculture/non-agriculture gap ratio of:
(A) 10.7, matching the data exactly
(B) 1.0, meaning selection explains nothing
(C) 2.2, roughly half of the 10.7 data target

(D) 45

Q8.4 Rich-country agricultural workers have average agricultural productivity relative to the population mean of about:

(A) 0.71
(B) 1.00
(C) 1.55, reflecting strong positive selection
(D) 10.7

Part V

Endogenous Agricultural Policy and Climate Shocks

This part is based on Hsiao, Moscona, and Sastry (2026, *Econometrica*), "Food Policy in a Warming World." The previous parts treat trade costs and productivity as the primitives driving reallocation. This part endogenizes a third margin: governments actively **choose** agricultural trade policy in response to climate shocks, and that policy choice redistributes the burden of climate damage — often regressively.

9 Lesson 9: Measuring Policy and Heat Exposure, and the Core Empirical Pattern

9.1 Data and key variables

Definition: Nominal Rate of Assistance and crop-specific extreme heat

The **Nominal Rate of Assistance** compares the distorted domestic producer price to the free-market international price,

$$NRA_{\ell kt} = \frac{p_{\ell kt} - p^I_{kt}}{p^I_{kt}},$$

with $NRA > 0$ signaling producer assistance and $NRA < 0$ signaling consumer assistance (e.g. export taxes, price controls). **Extreme heat exposure** is measured only on land actually growing crop k, using crop-specific heat tolerance thresholds T^{max}_k:

$$ExtremeHeat_{\ell kt} = \sum_{c \in \ell} \frac{Area_{ck}}{\sum_{c' \in \ell} Area_{c'k}} \cdot DegreeDays_{ct}(T^{max}_k).$$

This crop-specific construction is what allows identification from *within-country, across-crop* variation in heat exposure over time.

Derivation: Validating the heat measure

Before studying policy, the paper validates that the heat measure captures real production shocks: $\log(yield_{\ell kt}) = f(ExtremeHeat_{\ell kt}) + \gamma_{\ell t} + \delta_{kt} + \mu_{\ell k} + \varepsilon_{\ell kt}$, with country-year, crop-year, and country-crop fixed effects. Top-quartile heat exposure substantially lowers yields, confirming the heat measure is a valid production shock — the same logic as the FAO-GAEZ potential-yield validation exercises in Parts I–II.

9.2 Domestic vs. foreign shocks: opposite policy responses

Derivation: The baseline and foreign-shock specifications

$$NRA_{\ell kt} = g(ExtremeHeat_{\ell kt}) + \gamma_{\ell t} + \delta_{kt} + \mu_{\ell k} + \varepsilon_{\ell kt},$$

with $\gamma_{\ell t}$ (country-year), δ_{kt} (crop-year), and $\mu_{\ell k}$ (country-crop) fixed effects absorbing everything except the crop-specific heat shock. Foreign shocks enter analogously through leave-one-out global heat or import/export-trade-weighted heat, $ForeignExtremeHeat^{M}_{\ell kt} = \sum_{\ell' \neq \ell} ImportShare_{\ell' \ell k} \cdot ExtremeHeat_{\ell' kt}$.

Empirical Fact: The central empirical result

Domestic extreme heat lowers NRA (more pro-consumer policy: export restrictions, price controls to keep food affordable). **Foreign** extreme heat *raises* NRA (more pro-producer policy: less restriction on exports when domestic producers face a favorable world price). The response is concentrated in **border policies** (tariffs, export restrictions) rather than farm-gate or input policy, is **persistent** over several years, and is markedly stronger for **staple crops** and in **election years**.

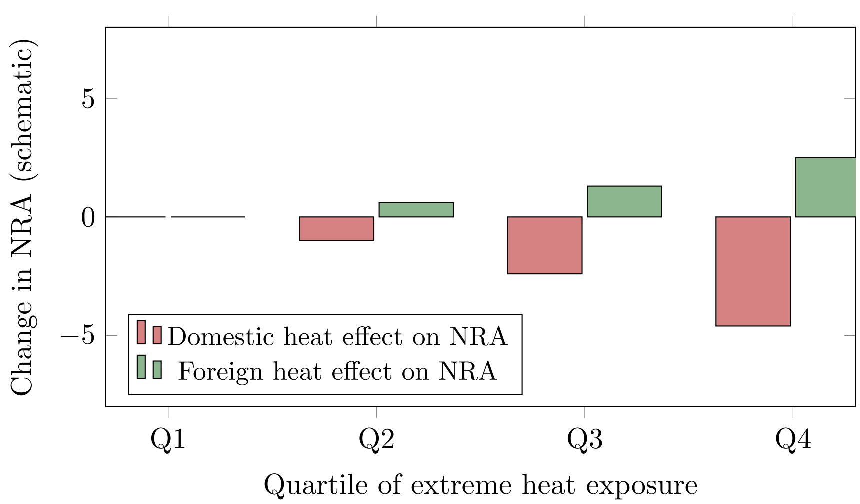

Figure 7: Schematic version of Hsiao-Moscona-Sastry's core finding: domestic and foreign extreme-heat shocks move agricultural policy (NRA) in *opposite* directions, consistent with a government redistributing between domestic consumers and producers.

Understanding Quiz 9 — Lesson 9

Q9.1 The Nominal Rate of Assistance $NRA < 0$ signals:
(A) producer assistance
(B) consumer assistance (domestic price below the free-market international price)
(C) no distortion
(D) a tariff on imports

Q9.2 $ExtremeHeat_{\ell kt}$ is constructed using crop-specific heat tolerance and land shares because:
(A) it allows identification from within-country, across-crop variation
(B) all crops have identical heat tolerance
(C) it ignores where crops are actually grown
(D) it is unrelated to yields

Q9.3 Domestic extreme heat and foreign extreme heat move NRA:
(A) in the same direction always
(B) in opposite directions: domestic heat lowers NRA (pro-consumer), foreign heat raises NRA (pro-producer)
(C) only affects input policy
(D) have no effect

Q9.4 The policy response to extreme heat is strongest for:
(A) cash crops in non-election years
(B) staple crops and in election years
(C) countries with no trade
(D) manufacturing goods

10 Lesson 10: A Political-Economy Model of Trade Policy, and Welfare

10.1 A tractable partial-equilibrium model

Definition: Demand, supply, and the border instrument

Consumer demand $q = Q(p) = p^{-\varepsilon_d}$, domestic supply $y = Y(p, \omega) = \omega p^{\varepsilon_s}$ (domestic productivity shock ω), and net export demand from the rest of the world $x = X(p, \omega') = \omega' p^{-\varepsilon_x}$ (foreign shifter ω'), with $\varepsilon_d < \varepsilon_x < \infty$. The government sets an ad valorem border tax $\alpha > -1$ linking the domestic and international price, $p^* = (1+\alpha)p^I$ ($\alpha > 0$: export subsidy; $\alpha < 0$: export tax).

Derivation: The government's objective and optimal policy

The government chooses α to maximize a weighted sum of consumer surplus, producer surplus, and (minus) fiscal cost,

$$\lambda^C \int_{p^*}^{\bar{p}} Q(p)\,dp + \lambda^P \int_0^{p^*} Y(p,\omega)\,dp - \lambda^G \frac{\alpha}{1+\alpha} p^* X\Big(\frac{p^*}{1+\alpha}, \omega'\Big),$$

subject to market clearing $p^* = P^*(\alpha, \omega, \omega')$. In terms of the **self-sufficiency ratio** $r = y/q$, the solution has closed form:

$$\alpha^* = \frac{1}{\varepsilon_x}\left(\frac{\varepsilon_x\big(\lambda^P r + \lambda^G(1-r) - \lambda^C\big) - \lambda^G(\varepsilon_s r + \varepsilon_d)}{\lambda^G(\varepsilon_s r + \varepsilon_d) - \big(\lambda^P r + \lambda^G(1-r) - \lambda^C\big)}\right).$$

Under utilitarian weights $\lambda^P = \lambda^C = \lambda^G$, this collapses to the inverse-elasticity, pure terms-of-trade benchmark $\alpha^* = -1/\varepsilon_x$: policy is about terms-of-trade manipulation, not shock response, *unless* political weights differ.

Derivation: Redistribution-focused vs. revenue-focused governments

Define the government as **redistribution-focused** if

$$\frac{\varepsilon_s\lambda^C + \varepsilon_d\lambda^P}{\varepsilon_s + \varepsilon_d} > \lambda^G,$$

and **revenue-focused** otherwise. If redistribution-focused, α^* *increases* in both ω and ω' (protecting whichever group the shock threatens); if revenue-focused, α^* *decreases* in both. A domestic adverse supply shock is a low-ω shock (lowers $r = y/q$); a foreign adverse supply shock is a high-ω' shock (raises world demand pressure). For a redistribution-focused government these two shocks push α^* in *opposite* directions — exactly matching the empirical pattern of Lesson 9.

10.2 Quantification: policy as redistribution, not neutral adaptation

Empirical Fact: Welfare consequences

Simulating the estimated policy response against a counterfactual of unresponsive (fixed) policy: in shocked markets, responsive policy dampens the price increase, *lowering* consumer losses but *raising* producer losses; in unshocked markets, spillovers can raise prices and redistribute losses elsewhere. Aggregate welfare can **worsen** even as consumers are protected, and poorer, more heat-exposed countries are harmed disproportionately — the paper's headline normative claim is that endogenous food policy is **globally regressive** even though it is domestically politically rational.

Intuition: The deep link to the rest of the module

This part shows that comparative advantage and technology (Parts I–II), sectoral composition (Part III), and worker selection (Part IV) are not the only forces shaping outcomes under climate change: the trade-policy *instrument itself* is an endogenous, politically determined object that reallocates the burden of shocks across consumers, producers, and foreigners. It sets up Part VI (Nath), which studies what happens to sectoral reallocation and welfare precisely *because* trade costs (including policy-induced ones) remain high in poor countries.

Understanding Quiz 10 — Lesson 10

Q10.1 Under utilitarian weights $\lambda^P = \lambda^C = \lambda^G$, optimal policy α^* reduces to:
(A) zero
(B) $-1/\varepsilon_x$, the pure terms-of-trade manipulation benchmark
(C) infinity
(D) $1/\varepsilon_d$

Q10.2 For a redistribution-focused government, a domestic adverse supply shock (low ω) and a foreign adverse supply shock (high ω'):
(A) both raise α^*
(B) both lower α^*
(C) push α^* in opposite directions
(D) have no effect on α^*

Q10.3 The paper's normative headline is that endogenous food policy:
(A) is always welfare-improving globally
(B) can be globally regressive even though it is domestically politically rational
(C) eliminates all climate damage
(D) only affects producers

Q10.4 A government is "redistribution-focused" when:
(A) λ^G dominates the weighted elasticity average
(B) the weighted average $\frac{\varepsilon_s \lambda^C + \varepsilon_d \lambda^P}{\varepsilon_s + \varepsilon_d}$ exceeds λ^G
(C) $\varepsilon_x = 0$
(D) there is no trade

Part VI

The Food Problem: Climate Change and Adaptation through Sectoral Reallocation

This final part is based on Nath (2025, *JPE*), "Climate Change, the Food Problem, and the Challenge of Adaptation through Sectoral Reallocation." It is the natural capstone of the module: it combines non-homothetic demand (Part III), a full Eaton-Kortum trade structure (as in the companion volume), and endogenous sectoral reallocation (Part IV) into a single general-equilibrium answer to the question posed by every earlier part — *can trade and reallocation save poor, hot countries from climate damage, or does the same subsistence-and-high-trade-cost combination that produces Lagakos-Waugh's selection amplification also trap poor countries in a worsening "food problem"?*

11 Lesson 11: Three Facts and the Non-Homothetic CES + Eaton-Kortum Model

11.1 Motivation and three empirical facts

Empirical Fact: Fact 1 — heat hurts agriculture far more

Using firm-level data from 17 countries, a 40°C day reduces annual output per worker by 0.4% in a poor, temperate country's agriculture, but is near-zero in rich countries and in non-agriculture generally. By 2100, projected warming reduces agricultural productivity by 14–21% on average versus only about 1.7% in manufacturing.

Empirical Fact: Fact 2 — poor countries specialize despite low relative productivity

Agriculture's GDP share is 67% in the poorest decile of countries versus 3% in the richest, even though relative value added per worker in agriculture is **45 times lower** in poor countries — the Lagakos-Waugh gap from Part IV, now embedded in a trade model.

Empirical Fact: Fact 3 — extreme heat raises agriculture's share

Country-level panel regressions show harmful temperature shocks *increase*, not decrease, the agricultural share of output and employment — the opposite of what a simple reallocation-out-of-damaged-sectors intuition would suggest.

Intuition: The food problem in one sentence

If trade were frictionless, a country hit by an agricultural productivity shock could simply import more food and shift workers into manufacturing — climate damage would be small. But if trade costs are high and food is a necessity (non-homothetic demand), falling agricultural productivity makes the country *poorer*, and being poorer makes it spend an even *larger* budget share on food, pulling *more* labor into the very sector that climate change is damaging. This vicious circle is the **food problem**, and it is precisely the poorest, most trade-restricted countries — Lagakos-Waugh's high-π_a, low-selection countries — that are most exposed to it.

11.2 Non-homothetic CES demand

Definition: Implicit CES utility and expenditure shares

Country k's consumer has implicit utility

$$\Omega_a^{1/\sigma} U_k^{\frac{\epsilon_a - \sigma}{\sigma}} C_{ak}^{\frac{\sigma-1}{\sigma}} + \Omega_m^{1/\sigma} U_k^{\frac{\epsilon_m - \sigma}{\sigma}} C_{mk}^{\frac{\sigma-1}{\sigma}} + \Omega_s^{1/\sigma} U_k^{\frac{\epsilon_s - \sigma}{\sigma}} C_{sk}^{\frac{\sigma-1}{\sigma}} = 1,$$

with $\sigma = 0.27 < 1$ (low cross-sector substitutability, estimated) and $\epsilon_a = 0.29 < 1 - \sigma$ (food is a necessity: utility elasticity below the substitution-adjusted threshold). By Roy's identity, the expenditure share on sector j is

$$\omega_{jk} = \Omega_j \left(\frac{P_{jk}}{P_k}\right)^{1-\sigma} \left(\frac{Y_k}{P_k}\right)^{\epsilon_j-(1-\sigma)}.$$

Derivation: Why real-income shocks raise food's share

When real income Y_k/P_k falls and $\epsilon_a - (1-\sigma) < 0$ (true whenever $\epsilon_a < 1-\sigma$, i.e. food is a necessity), the exponent on Y_k/P_k in ω_{ak} is negative, so ω_{ak} *rises* as real income falls — the formal statement of "the poor spend more on food."

11.3 Production: Eaton-Kortum with Fréchet productivity

Derivation: EK machinery, exactly as in the companion volume

Each sector-country productivity draw is Fréchet, $F_{jk}(z) = \exp(-Z_{jk}z^{-\theta_j})$, with $\theta_a = 4.06$, $\theta_m = 4.63$ (Tombe 2015). The standard EK price index and trade-share formulas apply sector by sector:

$$P_{jk} = \gamma_j \Big[\sum_n Z_{jn}(\tau_{jnk}w_n)^{-\theta_j}\Big]^{-1/\theta_j}, \qquad \pi_{jnk} = \frac{Z_{jn}(\tau_{jnk}w_n)^{-\theta_j}}{\sum_m Z_{jm}(\tau_{jmk}w_m)^{-\theta_j}}.$$

Labor-market clearing plus $P_{jk}C_{jk} = \omega_{jk}Y_k$ delivers the **employment-share equation**, the key object of the paper's comparative statics:

$$l_{jk} \equiv \frac{L_{jk}}{L_k} = \pi_{jkk}\,\omega_{jk} + \sum_{n\neq k} \pi_{jkn}\,\omega_{jn}\frac{Y_n}{Y_k}.$$

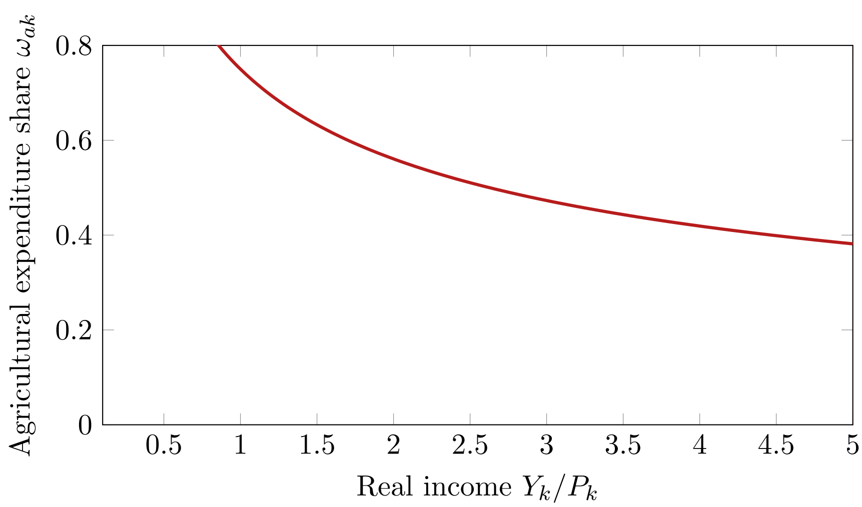

Figure 8: Non-homothetic CES: agricultural expenditure share falls as real income rises, since $\epsilon_a - (1-\sigma) < 0$. A negative real-income shock (e.g. from climate damage) moves a country back *up* this curve.

Understanding Quiz 11 — Lesson 11

Q11.1 Fact 3 (extreme heat raises agriculture's employment share) is surprising because naively one might expect:
(A) workers to flee the sector being damaged by heat, lowering its share
(B) no relationship between heat and employment shares
(C) heat to only affect manufacturing
(D) this is not actually surprising

Q11.2 In the non-homothetic CES system, $\epsilon_a < 1 - \sigma$ means:
(A) agriculture is a luxury
(B) agriculture is a necessity: its expenditure share falls as real income rises
(C) σ must equal 1
(D) there is no food problem possible

Q11.3 The employment-share equation $l_{jk} = \pi_{jkk}\omega_{jk} + \sum_{n\neq k} \pi_{jkn}\omega_{jn}Y_n/Y_k$ says a sector's labor share equals:
(A) only domestic consumption of the domestic good
(B) domestic consumption of the domestic good plus exports to other countries scaled by relative incomes
(C) a constant
(D) only exports

Q11.4 The trade elasticities used for agriculture and manufacturing are:
(A) $\theta_a = 4.06,\ \theta_m = 4.63$
(B) $\theta_a = 8.28$ (EK's estimate)
(C) both equal to σ
(D) unrelated to Fréchet dispersion

12 Lesson 12: The Food Problem Condition, Calibration, and Counterfactuals

12.1 Deriving the food-problem condition

Definition: Food problem

Country k has a food problem if an exogenous fall in agricultural productivity ($dZ_{ak} < 0$) *raises* its agricultural employment share ($dl_{ak} > 0$).

Derivation: Total differential of the employment-share equation

Differentiating $l_{ak} = \pi_{akk}\omega_{ak} + \sum_{n\neq k} \pi_{akn}\omega_{an}Y_n/Y_k$:

$$dl_{ak} = \underbrace{\pi_{akk}\, d\omega_{ak}}_{(A)} + \underbrace{\omega_{ak}\, d\pi_{akk}}_{(B)} + \underbrace{\sum_{n\neq k} \omega_{an}\, d\pi_{akn}\, \frac{Y_n}{Y_k}}_{(C)}.$$

Term (A) is positive: a fall in Z_{ak} raises P_{ak}/P_k (substitution effect, since $\sigma < 1$ means food cannot be easily replaced) and lowers real income (income effect, since $\epsilon_a < 1 - \sigma$) — both push ω_{ak} up. **Terms (B) and (C) are negative**: the country imports more of the now-scarcer good, so its domestic and export trade shares in agriculture fall ($d\pi_{akk} < 0$, $d\pi_{akn} < 0$), pulling l_{ak} down as trade substitutes for domestic production.

Derivation: The food-problem condition and two corollaries

A food problem occurs exactly when the positive expenditure-share effect dominates the negative trade-reallocation effects:

$$\boxed{\pi_{akk}\, d\omega_{ak} > -\Big(\omega_{ak}\, d\pi_{akk} + \sum_{n\neq k} \omega_{an}\, d\pi_{akn}\, \tfrac{Y_n}{Y_k}\Big)}\,.$$

Corollary 1 (frictionless trade): if $\tau = 1$ everywhere, $\pi_{akk} \to 0$ for a small country, term (A) vanishes, and $dl_{ak} < 0$ — no food problem. **Corollary 2 (homothetic preferences)**: if $\epsilon_j = 1 - \sigma$ for all j with $\sigma \geq 1$, term (A) is non-positive and again $dl_{ak} < 0$. **The food problem strictly requires both** high trade barriers (large π_{akk}) *and* non-homothetic, low-substitutability preferences — exactly the ingredients emphasized throughout this module (Part III's subsistence demand, Part IV's high poor-country agricultural employment share, Part V's high trade barriers sustained by policy).

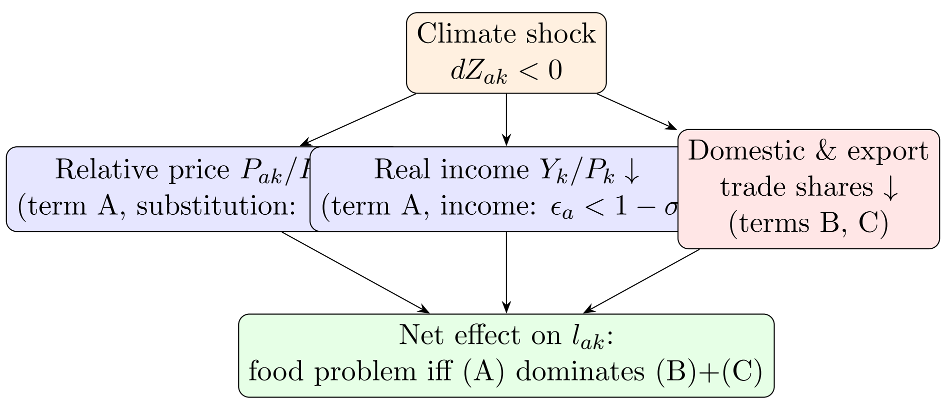

Figure 9: The anatomy of the food-problem condition: a negative agricultural productivity shock pushes the agricultural employment share up through relative-price and income channels (term A), and down through trade-reallocation channels (terms B, C). The food problem is precisely the case where high trade costs make (A) dominate.

12.2 Calibration and counterfactuals

Empirical Fact: Calibrated parameters (158 countries, 2011)

$\sigma = 0.27$, $\epsilon_a = 0.29$, $\epsilon_m = 1.00$, $\epsilon_s = 1.15$, $\theta_a = 4.06$, $\theta_m = 4.63$; trade costs calibrated to match bilateral Comtrade flows. The poorest quartile of countries imports only **9%** of its food ($\pi_{akk} = 0.91$); the richest quartile imports **55%**.

Empirical Fact: Headline counterfactual results

Under *current* trade costs, climate change **raises** agriculture's GDP share in the poorest quartile by **2.8 percentage points** — the food problem is quantitatively real — with a welfare loss for the poorest quartile of -6.6% of income. If trade costs were lowered to the 90th-percentile openness level, the welfare loss falls to -3.8% and agriculture's share *falls* rather than rises. Universal free-trade agreements and reduced red tape achieve roughly **one-third** of the maximum possible adaptation gains available to poor countries.

Intuition: The module's throughline, completed

Nath's result closes the circle opened in Part I: Costinot-Donaldson-Smith showed that *if* countries can reallocate crops and trading partners, climate damage is modest (a 0.26% world welfare loss). Nath shows that this optimistic view implicitly assumes low trade costs; once realistic trade barriers are combined with the subsistence, non-homothetic preferences of Part III and the same selection forces that generate Lagakos-Waugh's productivity gaps in Part IV, the poorest countries are pushed *further* into low-productivity agriculture rather than out of it. And Part V shows that the trade barriers sustaining this trap are not exogenous: they are, in part, policy choices that domestic political economy can make worse in exactly the moments adaptation is most needed. Lowering trade costs — both natural and policy-induced — is therefore the single lever that runs through every part of this module.

Understanding Quiz 12 — Lesson 12

Q12.1 The food-problem condition requires the positive term (A) to dominate terms (B) and (C), which requires:
- (A) low trade costs and homothetic preferences
- (B) high trade costs (large π_{akk}) and non-homothetic, low-substitutability preferences
- (C) $\theta_a = \theta_m$
- (D) free migration

Q12.2 Under frictionless trade ($\tau = 1$), term (A) vanishes for a small country because:
- (A) $\pi_{akk} \to 0$: the country barely produces its own food, so domestic price effects carry no weight in its overall employment share
- (B) $\sigma \to \infty$
- (C) $\epsilon_a \to 0$
- (D) wages become infinite

Q12.3 At current trade costs, climate change is estimated to change the poorest quartile's agricultural GDP share by:
- (A) -2.8 points
- (B) $+2.8$ points (a genuine food problem)
- (C) 0
- (D) $+28$ points

Q12.4 Moving to 90th-percentile trade openness:
- (A) worsens welfare losses to -10%
- (B) reduces the poorest quartile's welfare loss from -6.6% to -3.8% and reverses the rise in agriculture's share
- (C) has no effect

(D) only helps rich countries

Final Exam: Comprehensive Multiple-Choice Quiz

Understanding Quiz — Final Exam (all parts)

F1. In Costinot-Donaldson-Smith, a plot's land productivity draw is:
(A) a single national number
(B) a Fréchet-distributed random variable specific to that plot and crop
(C) always equal across crops
(D) irrelevant to output

F2. FAO-GAEZ "potential yield" should be interpreted as:
(A) the observed yield on a field
(B) the yield if the field were entirely devoted to one crop — an agro-climatic, not socio-economic, measure
(C) a price index
(D) a labor supply elasticity

F3. In CDS's counterfactual, freezing the *production* margin (crop switching) relative to full adjustment:
(A) roughly triples the world welfare loss from climate change
(B) eliminates all losses
(C) has no effect
(D) only matters for trade

F4. Farrokhi-Pellegrina's generalized Fréchet nests:
(A) countries within crops
(B) crops (dispersion θ_1) and, within each crop, technologies (dispersion θ_2)
(C) only technologies, no crops
(D) only labor, no land

F5. The Farrokhi-Pellegrina gains-from-trade formula $G_i = 1 - \lambda_{ii}^{1/(\sigma-1)} \alpha_{i,0}^{1/\theta_2}$ shows that the classical ACR formula alone:
(A) fully captures all gains from trade in agriculture
(B) misses gains from imported-input-driven technology adoption, captured by the second term
(C) is exactly equal to the technology term
(D) does not apply to agriculture

F6. Empirical Pattern 3 in Farrokhi-Pellegrina (large yield premia not varying systematically with income) implies that cross-country adoption gaps mostly reflect:
(A) agro-climatic potential differences
(B) economic incentives (prices, market access) rather than agronomy
(C) random noise
(D) labor supply

F7. In the Herrendorf-Rogerson-Valentinyi benchmark model, the key decomposition is:

(A) growth equals structural transformation exactly
(B) growth (intertemporal problem, driven by A_{xt}) is separate from structural transformation (static allocation problem)
(C) there is no growth in the model
(D) all sectors grow at the same rate always

F8. A generalized balanced growth path (GBGP) requires:
(A) constant sectoral shares
(B) a constant real interest rate while sectoral shares evolve freely
(C) zero technology growth
(D) constant relative prices

F9. The Ngai-Pissarides mechanism for structural transformation relies on:
(A) subsistence consumption alone
(B) differential sectoral TFP growth combined with low substitutability across sectors
(C) free trade
(D) labor mobility frictions

F10. Opening a structural-transformation economy to international trade:
(A) makes production and expenditure shares identical by construction
(B) breaks the tight closed-economy link between production and expenditure shares, since firms specialize by comparative advantage
(C) eliminates all sectors but one
(D) has no effect on manufacturing's hump shape

F11. Lagakos-Waugh's central empirical fact is that cross-country productivity gaps are:
(A) identical across agriculture and non-agriculture
(B) roughly ten times larger in agriculture (45×) than non-agriculture (4×)
(C) larger in non-agriculture
(D) zero in both sectors

F12. In the Lagakos-Waugh Roy model, a worker selects into non-agriculture when:
(A) $z_i^n/z_i^a \geq p_a$
(B) $z_i^a/z_i^n \geq p_a$
(C) wages are equal in both sectors regardless of type
(D) never

F13. Proposition 2 in Lagakos-Waugh states that positive sorting on comparative advantage:
(A) shrinks the agricultural productivity gap below the efficiency gap A_R/A_P
(B) amplifies the agricultural gap above A_R/A_P and shrinks the non-agricultural gap below it
(C) has no effect on measured gaps
(D) only affects non-agriculture

F14. The calibrated Lagakos-Waugh model explains roughly what share of the observed 10.7× agriculture/non-agriculture productivity-gap amplification?
(A) 0%
(B) about half (model ratio 2.2 vs. 1.0 with no selection, against a 10.7 target)
(C) 100% exactly
(D) over 200%

F15. In Hsiao-Moscona-Sastry, domestic extreme heat and foreign extreme heat move the Nominal Rate of Assistance (NRA):
(A) in the same direction

(B) in opposite directions — domestic heat lowers NRA (pro-consumer), foreign heat raises NRA (pro-producer)
(C) only through input policy
(D) not at all

F16. Under utilitarian political weights ($\lambda^P = \lambda^C = \lambda^G$), optimal border policy α^* in the Hsiao-Moscona-Sastry model:
(A) responds strongly to domestic and foreign shocks
(B) collapses to the pure terms-of-trade benchmark $-1/\varepsilon_x$, independent of shocks
(C) is always positive
(D) is undefined

F17. The paper's headline normative finding on endogenous food policy is that it is:
(A) always globally welfare-improving
(B) can be globally regressive even though domestically politically rational
(C) irrelevant to welfare
(D) beneficial only to foreign producers

F18. In Nath (2025), the food-problem condition requires:
(A) low trade costs and homothetic preferences
(B) high trade costs (large domestic trade share π_{akk}) together with non-homothetic, low-substitutability preferences
(C) $\sigma > 1$
(D) no agriculture sector

F19. Under frictionless trade, Nath's Corollary 1 shows that a small country's food problem:
(A) is maximized
(B) disappears, since the domestic-price term (A) vanishes as $\pi_{akk} \to 0$
(C) is unaffected
(D) only appears for rich countries

F20. At current (high) trade costs, Nath's calibration implies climate change
(A) lowers agriculture's GDP share in the poorest quartile of countries
(B) raises agriculture's GDP share in the poorest quartile by about 2.8 percentage points, a genuine food problem
(C) has zero effect on any country
(D) only affects rich countries

F21. What is the common mathematical thread linking the Roy-Fréchet selection models of Costinot-Donaldson-Smith, Farrokhi-Pellegrina, and Lagakos-Waugh?
(A) heterogeneous units (plots or workers) draw random productivities and select into the option with the highest return, and Fréchet dispersion parameters govern how responsive that selection is to relative-return changes
(B) they all assume homogeneous agents
(C) none of them use discrete choice
(D) they all study only manufacturing

F22. Which single policy lever appears, in one form or another, throughout all six papers in this module as the key determinant of how much climate/trade shocks damage welfare?
(A) the level of (natural and policy-induced) trade costs
(B) the interest rate
(C) the level of nominal wages
(D) the exchange rate regime

A Answer Key

Lesson quizzes

Lesson 1

Q1.1: B. Potential yield is the agro-climatic mean yield if the field were devoted entirely to that crop, not the observed (selected) yield. **Q1.2: B.** Production (crop switching) and trade (source-country switching) are the two adjustment margins. **Q1.3: A.** Plots are allocated to whichever option (crop or idle) delivers the highest realized return. **Q1.4: C.** Conditional mean productivity exceeds the unconditional mean because only favorable draws are planted, scaling as $(\pi^f_{i,k})^{-1/\theta}$.

Lesson 2

Q2.1: B. Potential yield is an agro-climatic supply shifter, plausibly excluded from demand shocks. **Q2.2: C.** Only (α_i, θ) remain to be estimated, matched to agricultural land and output. **Q2.3: B.** Freezing production adjustment roughly triples the world welfare loss. **Q2.4: B.** $\theta = 2.46$ for land is low relative to manufacturing trade elasticities, signaling strong comparative-advantage scope across crops.

Lesson 3

Q3.1: B. Similar potential everywhere implies adoption gaps reflect economic incentives, not agronomy. **Q3.2: B.** θ_1: across crops within a field; θ_2: across technologies within a crop. **Q3.3: B.** $\gamma^M_{k0} = 0$ vs. $\gamma^M_{k1} > 0$ distinguishes traditional from modern technology. **Q3.4: A.** Distance to hubs proxies market access, holding potential yield fixed.

Lesson 4

Q4.1: B. The $\alpha_{i,0}^{1/\theta_2}$ term captures technology-adoption gains from imported inputs. **Q4.2: D.** The modern-share-vs.-trade-cost moment is most informative about θ_2. **Q4.3: B.** Reversing input trade costs alone lowers modern-technology adoption, yields, and welfare by about 1.6%. **Q4.4: B.** Output globalization benefits poor countries (high food expenditure share); input globalization benefits middle-income countries (intermediate room to adopt).

Lesson 5

Q5.1: B. The three measures are conceptually distinct and can diverge with trade or IO linkages. **Q5.2: C.** Manufacturing is hump-shaped over the growth path. **Q5.3: A.** Growth (intertemporal, via A_{xt}) is separate from structural transformation (static allocation). **Q5.4: B.** GBGP requires a constant real interest rate, not constant sectoral shares.

Lesson 6

Q6.1: B. Kongsamut-Rey-Xie relies on non-homothetic preferences with roughly common TFP growth. **Q6.2: B.** Ngai-Pissarides relies on differential TFP growth plus low substitutability. **Q6.3: A.** Trade lets production specialize by comparative advantage, independent of domestic consumption patterns. **Q6.4: B.** Strong manufacturing TFP growth plus trade can raise the manufacturing production share via specialization even as consumption shifts to services.

Lesson 7

Q7.1: B. The puzzle is the roughly ten-times-larger agricultural productivity gap. **Q7.2: A.** A worker chooses non-agriculture when $z_i^n/z_i^a \geq p_a$. **Q7.3: B.** Poor countries have a higher relative price of agriculture, since subsistence demand binds more tightly. **Q7.4: B.** Measured productivity depends on both common efficiency A and selection.

Lesson 8

Q8.1: B. Selection amplifies the agricultural gap above A_R/A_P and shrinks the non-agricultural gap below it. **Q8.2: B.** A falling π_a mechanically raises selected agricultural productivity. **Q8.3: C.** The calibrated model delivers a ratio of 2.2, versus a 10.7 data target and 1.0 with no selection. **Q8.4: C.** Rich-country agricultural workers average about 1.55× the population mean productivity.

Lesson 9

Q9.1: B. $NRA < 0$ signals consumer assistance (domestic price below international price). **Q9.2: A.** Crop-specific construction enables within-country, across-crop identification. **Q9.3: B.** Domestic heat lowers NRA; foreign heat raises NRA — opposite directions. **Q9.4: B.** The response is strongest for staple crops and in election years.

Lesson 10

Q10.1: B. Utilitarian weights collapse policy to the pure terms-of-trade benchmark $-1/\varepsilon_x$. **Q10.2: C.** For a redistribution-focused government, domestic and foreign shocks push α^* in opposite directions. **Q10.3: B.** Endogenous food policy can be globally regressive despite being domestically rational. **Q10.4: B.** Redistribution-focused means the weighted elasticity average exceeds λ^G.

Lesson 11

Q11.1: A. Naively one would expect workers to flee the damaged sector, making Fact 3 surprising. **Q11.2: B.** $\epsilon_a < 1 - \sigma$ means agriculture is a necessity. **Q11.3: B.** Employment share equals domestic consumption of domestic goods plus income-weighted exports. **Q11.4: A.** $\theta_a = 4.06$, $\theta_m = 4.63$ (Tombe 2015 estimates used by Nath).

Lesson 12

Q12.1: B. The food problem needs high trade costs plus non-homothetic, low-substitutability preferences. **Q12.2: A.** $\pi_{akk} \to 0$ for a small open economy, so term (A) vanishes. **Q12.3: B.** Climate change raises the poorest quartile's agricultural GDP share by about 2.8 points at current trade costs. **Q12.4: B.** Greater openness cuts the welfare loss from -6.6% to -3.8% and reverses the rise in agriculture's share.

Final exam

Final Exam key

F1: B. Plot-level Fréchet productivity draws, specific to plot and crop.
F2: B. Potential yield is an agro-climatic, not socio-economic, construct.
F3: A. Freezing production adjustment roughly triples the world welfare loss.
F4: B. Generalized Fréchet nests crops (θ_1) and, within crops, technologies (θ_2).
F5: B. The ACR term alone misses input-driven technology-adoption gains.
F6: B. Similar potential everywhere implies adoption gaps reflect economic incentives.
F7: B. Growth (intertemporal) is distinct from structural transformation (static allocation).
F8: B. GBGP requires a constant real interest rate, not constant shares.
F9: B. Ngai-Pissarides relies on differential TFP growth plus low substitutability.
F10: B. Trade breaks the closed-economy link between production and expenditure shares.
F11: B. Agricultural gaps (≈45×) are roughly ten times non-agricultural gaps (≈4×).
F12: A. A worker selects non-agriculture when $z_i^n/z_i^a \geq p_a$.
F13: B. Positive sorting amplifies the agricultural gap and shrinks the non-agricultural gap.
F14: B. The model explains roughly half of the observed amplification (2.2 vs. 10.7, with 1.0 as the no-selection benchmark).
F15: B. Domestic and foreign heat move NRA in opposite directions.
F16: B. Utilitarian weights collapse policy to the pure terms-of-trade benchmark, independent of shocks.
F17: B. Endogenous food policy can be globally regressive despite domestic political rationality.
F18: B. The food problem needs high trade costs plus non-homothetic, low-substitutability preferences.
F19: B. Under frictionless trade the food problem disappears for a small country.
F20: B. At current trade costs, climate change raises the poorest quartile's agricultural GDP share by about 2.8 points.
F21: A. Random productivity draws plus selection into the highest-return option, governed by a Fréchet dispersion parameter, is the common thread.
F22: A. The level of natural and policy-induced trade costs is the recurring determinant of adaptation and welfare throughout the module.